

Wyatt Walsh

SENIOR AI/ML ENGINEER

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Senior AI/ML engineer shipping production agentic AI, LLM document intelligence, retrieval workflows, AI guardrails, data platforms, and fintech tooling.

EXPERIENCE

Senior AI/ML Engineer

Frame Payments | Remote | Oct 2025 - Jan 2026

- Delivered production agentic AI document intelligence, human-in-the-loop review, and multimodal compliance screening for high risk merchant onboarding.
- Deployed production LLM document intelligence with PydanticAI, Amazon Bedrock AgentCore, and Claude; added human-in-the-loop browser feedback for confidential uploads and reduced analyst follow-ups/resubmissions ~90% by blocking miscategorized documents before review.
 - Built multimodal restricted-item compliance screening by combining product images and structured metadata; translated banking-partner policy requirements into automated risk checks for merchant onboarding across tens of merchants and multi-location catalogs.
 - Built a NotebookLM-based technical knowledge hub spanning vendor/web documentation and internal monorepo docs; automated corpus refresh with a Colab pipeline that re-ingests the latest sources to keep retrieval accurate and up to date.
 - Drove internal AI adoption with Notion MCP playbooks, high-signal prompt patterns, and weekly office hours reaching ~15% of the company; standardized PII/confidential-document safeguards and auditability requirements.

Associate III, Senior Software Engineer - Data Integrations

JPMorgan Chase & Co. | New York, NY | May 2023 - Apr 2025

- Built enterprise data pipelines, risk analytics, and data-quality automation for credit trading while scaling secure internal AI developer tooling.
- Pioneered firm-wide rollout of secure generative AI developer tools (LLMs, GitHub Copilot, Codeium) as one of the first 500 engineers, shaping scalable deployment and leading workshops for technical and non-technical teams.
 - Devised ML-driven anomaly detection for credit trading and risk analytics, reducing false positives by 25% and manual review time for risk teams.
 - Developed a resilient Python wrapper for KDB+ with CI-style tests and optimized Q functions, democratizing data access for 5,000+ users.
 - Engineered a cloud-function-driven asset-fee data sink, consolidating 10+ sources into a centralized database, cutting manual maintenance by 90% and providing on-demand access for traders through a Python SDK, REST API, and interactive dashboard.

Senior Software Engineer

Stealth Web3 Startup | Remote | Mar 2022 - Jul 2022

- Architected a scalable collateralized lending app with React, Next.js, TypeScript, MobX, and ethers.js for secure wallet and smart-contract workflows.
- Implemented CI/CD with GitHub Actions, Husky, Cypress, and Jest, halving release cycles, raising test coverage to 90%, and mentoring a front-end intern.
- Optimized front-end performance through code splitting, lazy loading, and asset caching, reducing initial load times by 40% and JavaScript bundle size by 30%.

Founder, Chief Technology Officer

SandLabs | Remote | Jun 2021 - Mar 2022

- Launched a Next.js/TailwindCSS public site with top-tier Lighthouse scores and secured \$75K in competitive ecosystem grant funding.
- Built Python pipelines that normalized a 1M+ Reddit posts/comments dataset into a reusable analytics data product for external users.

Demand Forecasting Extern

Gap Inc. | San Francisco, CA | Jan 2020 - May 2020

- Applied time-series decomposition and bias correction techniques to demand forecasts, boosting accuracy by up to 25% and minimizing surplus inventory costs.
- Owned the data engineering layer by architecting scalable MySQL and Apache Spark pipelines across 5+ years of sales data, improving data processing throughput by 95% and standardizing insights across forecasting teams.

SKILLS

- MACHINE LEARNING AND GENERATIVE AI:** scikit-learn, Hugging Face Transformers, OpenAI API, LangChain, LangGraph, Model Context Protocol (MCP), Retrieval Augmented Generation (RAG), Prompt Engineering, PydanticAI, Amazon Bedrock, Claude, GitHub Copilot, Document AI, AI Guardrails
- DATA ENGINEERING AND ANALYTICS:** Pandas, NumPy, SQLAlchemy, Pydantic, Pandera, Apache Spark, AWS Glue, KDB+, PostgreSQL, DynamoDB, ETL, DuckDB, Polars, Parquet
- PROGRAMMING & SCRIPTING LANGUAGES:** Python, SQL, TypeScript, JavaScript, Go, Rust, Bash
- CLOUD INFRASTRUCTURE AND DEVOPS:** AWS, Docker, Kubernetes, Terraform, GitHub Actions, AWS Lambda, Google Cloud Functions, ECS, EKS

PROJECTS

AI Agent Harness Configs | github.com/wyattowalsh/agents

Python/TypeScript MCP developer-workflow catalog with 40+ reusable skills, FastMCP configs, shared instructions, and cross-runtime agent automation.

- Built 40+ instruction and skill packs plus harness configs across Claude Code, Codex, GitHub Copilot, Cursor, and related IDE agents.
- Shipped generated docs, install flows, and portable tool and MCP presets for code review, security, infrastructure, research, and workflow automation.

NBA Basketball Database | github.com/wyattowalsh/nbadb

Python warehouse using Pandas, Polars, SQLAlchemy, DuckDB, and Pandera over 65K+ NBA games, 13M+ events, Parquet exports, docs, and CLI.

- Implemented Pandera-validated NBA API ETL with daily warehouse updates plus DuckDB, SQLite, Parquet, and CSV exports.
- Built docs, schema references, release workflows, and Kaggle distribution for a 180+ table public warehouse spanning 130+ stats.nba.com endpoint classes.

Listentropy | github.com/wyattowalsh/listentropy

TypeScript/React listening-analytics app using Web Workers, Canvas exports, and Vercel to process private Spotify history in-browser.

- Architected a Web Worker processing pipeline with stage-level progress tracking that handles full streaming history datasets in-browser, keeping all personal listening data local.

ProxyWhirl | github.com/wyattowalsh/proxywhirl

Python/SQLite/PyPI data-collection toolkit with nine rotation strategies, 100+ auto-validated sources, and self-healing proxy health checks.

- Implemented 9 proxy rotation strategies with configurable fallback chains, per-source scoring, and performance-based selection.
- Built self-healing validation for 100+ proxy sources with circuit breakers, auto-ejection, and recovery promotion.

AI/ML Web Feeds | github.com/wyattowalsh/ai-web-feeds

Python source-discovery platform with Pydantic, SQLAlchemy, YAML feeds, Fumadocs docs, Vercel deploys, HTTP/RSS validation, and OPML exports.

- Defined YAML feed schemas with topic and source-type validation across blogs, papers, and newsletters for reliable feed quality.
- Shipped helper scripts, CI checks, scheduled validation jobs, and live docs for reliable community feed review.

MCP-Crawl4AI | github.com/wyattowalsh/mcp-crawl4ai

Python/FastMCP server using Crawl4AI, Playwright, and Docker so AI agents can crawl, extract, and return structured web content over stdio/HTTP.

- Built a FastMCP v3 server with managed headless Chromium, session-aware crawling, and automatic Playwright setup using two runtime dependencies.
- Exposed stdio/HTTP tools, resources, and prompt templates with docs, tests, and release-tagged MCP install flows.

MDXPad | github.com/wyattowalsh/mdxpad

Electron/React/TypeScript macOS editor with CodeMirror, Vercel AI SDK assistance, synchronized MDX preview, and Keychain credentials.

- Built a dual-format editor on Electron 39 + CodeMirror 6 with synchronized scroll between source and rendered preview, remark/rehype plugin support, and YAML frontmatter validation.

Personal Website: w4w.dev | github.com/wyattowalsh/personal-website

Next.js/React/TypeScript personal web app using Tailwind CSS and MDX for durable content, SEO, polished UI, and authoring workflows.

- Maintained the public site across multiple generations, covering responsive landing pages, themes, motion, accessibility, and MDX publishing.
- Built custom MDX publishing with search, metadata, math rendering, feeds, tags, and integrated authoring tools.

EDUCATION & CERTIFICATIONS

Bachelor of Science

University of California, Berkeley - College of Engineering, Industrial Engineering and Operations Research (IEOR)
Aug 2014 - Dec 2020

High School Diploma

The Hotchkiss School
Aug 2012 - May 2014

AWS Certified Cloud Practitioner

Amazon Web Services (AWS) | Nov 2024

IBM Data Science Professional Certificate

IBM | Feb 2021